ISSN 1028-334X, Doklady Earth Sciences, 2018, Vol. 482, Part 1, pp. 1134–1137. © Pleiades Publishing, Ltd., 2018.
Original Russian Text © I.V. Golovanova, K.N. Danukalov, V.N. Puchkov, A.M. Kosarev, R.Yu. Salmanova, 2018, published in Doklady Akademii Nauk, 2018, Vol. 482, No. 1.

GEOLOGY

Paleomagnetic Study of Devonian and Carboniferous Rocks of the Southern Urals: An Independent Test of Collision between the Magnitogorsk Island Arc and the Passive Margin of the Continent of Laurussia

I. V. Golovanova[a,*], K. N. Danukalov[a], Corresponding Member of the RAS V. N. Puchkov[a], A. M. Kosarev[a], and R. Yu. Salmanova[a]

Received December 20, 2017

Abstract—New paleomagnetic data on Devonian and Lower Carboniferous rocks of the Magnitogorsk Zone (Southern Urals) are presented. The paleomagnetic pole calculated for the Lower–Middle Devonian is quite close to the coeval pole of the Baltica Paleocontinent (Laurussia), which indicates that the Magnitogorsk island arc was probably located close to the continent, but not being accreted with it. Paleomagnetic data obtained for Lower Carboniferous rocks may indicate that the Magnitogorsk island arc was turned in the Early Carboniferous and collided with the continent of Laurussia forming a single continent.

DOI: 10.1134/S1028334X18090052

According to the current ideas ([6] and references therein), the continent of Laurussia and the Magnitogorsk island arc belonged to adjacent lithospheric plates in the Devonian time, being separated by a oceanic crust band and a subduction zone dipping to the east (in modern coordinates). In the Famennian, the subduction process led to an oblique (with a sinistral component) continent–island arc collision. In the course of the collision, the Magnitogorsk arc was accreted to the continental margin. Paleomagnetic data could serve as an independent way of estimating paleotectonic reconstructions. However, the amount of data available and especially their reliability are insufficient.

Since the scarce paleomagnetic Devonian age data available for the Magnitogorsk zone and the western slope of the Southern Urals [1, 2, 5, 7] were obtained before 1990, they do not meet the current methodological requirements or are contradictory. There are no published reliable paleomagnetic data on the Lower Carboniferous rocks of the Magnitogorsk zone of the Southern Urals. This work presents new paleomagnetic data on the Devonian and Lower Carboniferous rocks from the western and central parts of the Magnitogorsk zone, allowing one to compare the coeval paleolatitudes of the Magnitogorsk island arc and the eastern edge of the East European Craton and determine the mutual displacements of these two tectonic structures.

The sections of Devonian and Lower Carboniferous volcanic and volcanic–sedimentary rocks of the Magnitogorsk zone of the Southern Urals were studied (Fig. 1). In total, more than 1300 oriented samples were collected for paleomagnetic studies. The paleomagnetic data obtained for 12 Lower–Middle Devonian and 8 Lower Carboniferous sections were interpreted.

Samples of Devonian rocks were collected in the West Magnitogorsk Zone in a band extending about 250 km from north to south. All studied sequences were dated based on the finds of conodonts [4]. The paleomagnetic data were obtained for the Lower–Middle Devonian rocks ($D_1e_2 - D_2ef_2$). The rocks studied were primarily represented by mafic volcanics and red jasperoids; there are felsic volcanics and tuff sandstones, as well.

A number of sections of the Central Magnitogorsk zone are located within the Magnitogorsk–Bogdanovka graben. Geological sections of the mafic and felsic lavas (C_1) have been sampled at the sites located in the Ural River basin from the mouth of the Grekhovka River to the mouth of the Karaganka River.

The analytical processing of samples was performed at the Institute of Geology, Ufa Scientific Center, Russian Academy of Sciences (IG USC RAS, Ufa) and at the Geological Institute, Russian Academy of Sciences (GIN RAS, Moscow). All samples

[a] *Institute of Geology, Ufa Scientific Center, Russian Academy of Sciences, Ufa, 450077 Russia*
**e-mail: golovanova@ufaras.ru*

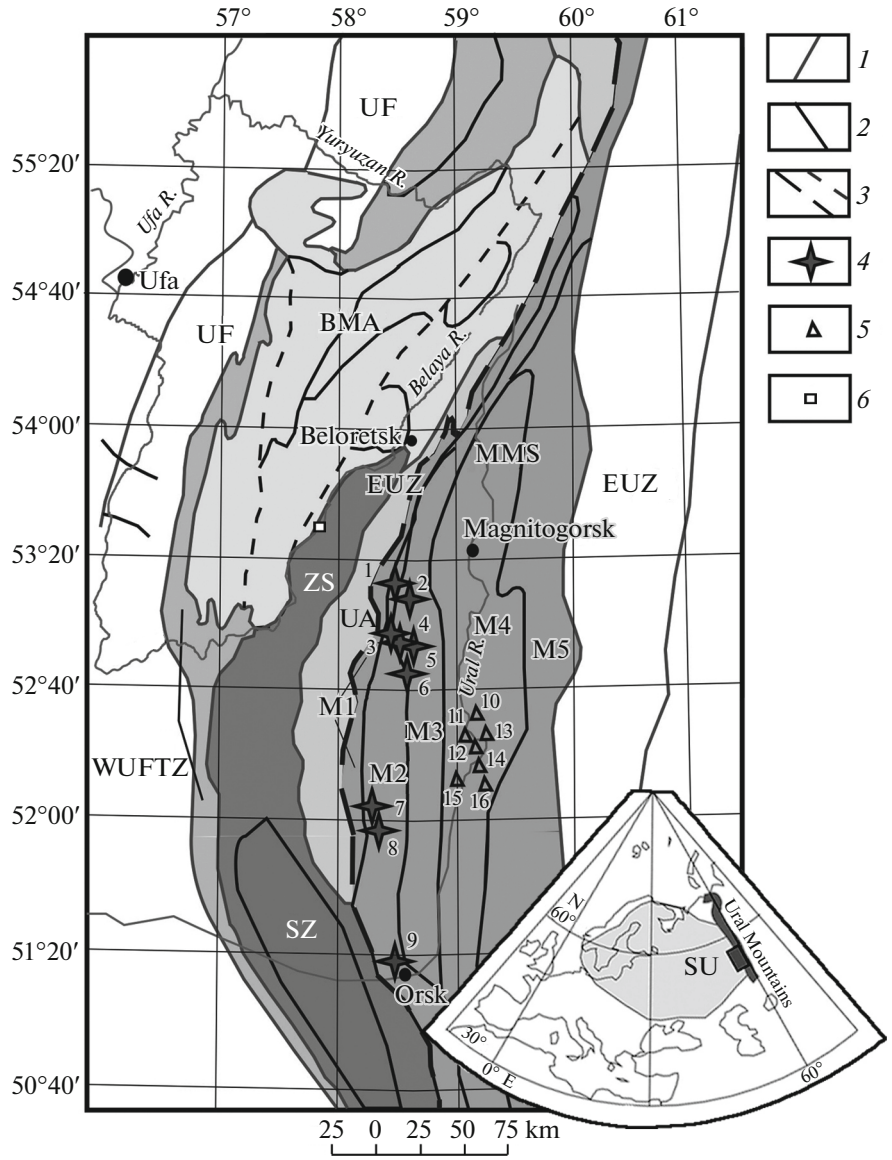


Fig. 1. Simplified structural–tectonic map of the Southern Urals showing sampling sites. Inset: the position of the Baltica paleocontinent with the Precambrian basement (shown in gray), the Urals, and the study area (SU). *1*, Boundary between the first-order structures; *2*, boundary between the second-order structures; *3*, major tectonic faults. The Main Ural Fault is shown by a thick dashed line. Sampled sections: *4*, Devonian; *5*, Carboniferous; *6*, sampling site Kaga; the paleomagnetic directions for this site were recalculated using the data from [7]. Abbreviations in the map: UF—Uralian Foredeep; WUFTZ—West Uralian Fold and Thrust Zone; BMA—Bashkir Meganticlinorium; ZS—Zilair Synclinorium; UA—Uraltau anticlinorium; MMS—Magnitogorsk Megasynclinorium; M1—Voznesenka–Sakmara Zone; M2, M3—West Magnitogorsk Zone: M2—Irendyk subzone; M3—Kizil–Urtazym subzone; M4—Central Magnitogorsk Zone; M5—East Magnitogorsk Zone; EUZ—East Uralian Zone.

were subject to detailed magnetic thermal cleaning. The remanent magnetization was measured on the JR-6 and JR-4 spin-magnetometers.

The absolute value of the natural remnant magnetization varies over a wide range, but, in general, volcanics and red jasperoids are characterized by high values. The high-temperature magnetization component, sometimes bipolar, was established in samples of mafic volcanics from most of the sections studied and in jasperoids (Fig. 2). Usually, it points to the origin of coordinates. It is not uncommon that this component is isolated above 580°C. High-temperature magnetization components are more often associated with magnetite, sometimes with hematite. The reversal test is positive both for the Devonian samples (γ/γ_c = 9.2/18.4, class C) and for the Lower Carboniferous (γ/γ_c = 9.9/19.4, class C). The mean directions of the high-temperature component measured with respect to all sampling sites for each group of sections are much better grouped in the ancient coordinate system.

In the sections of coeval rocks studied on the eastern margin of the East European Craton, primary

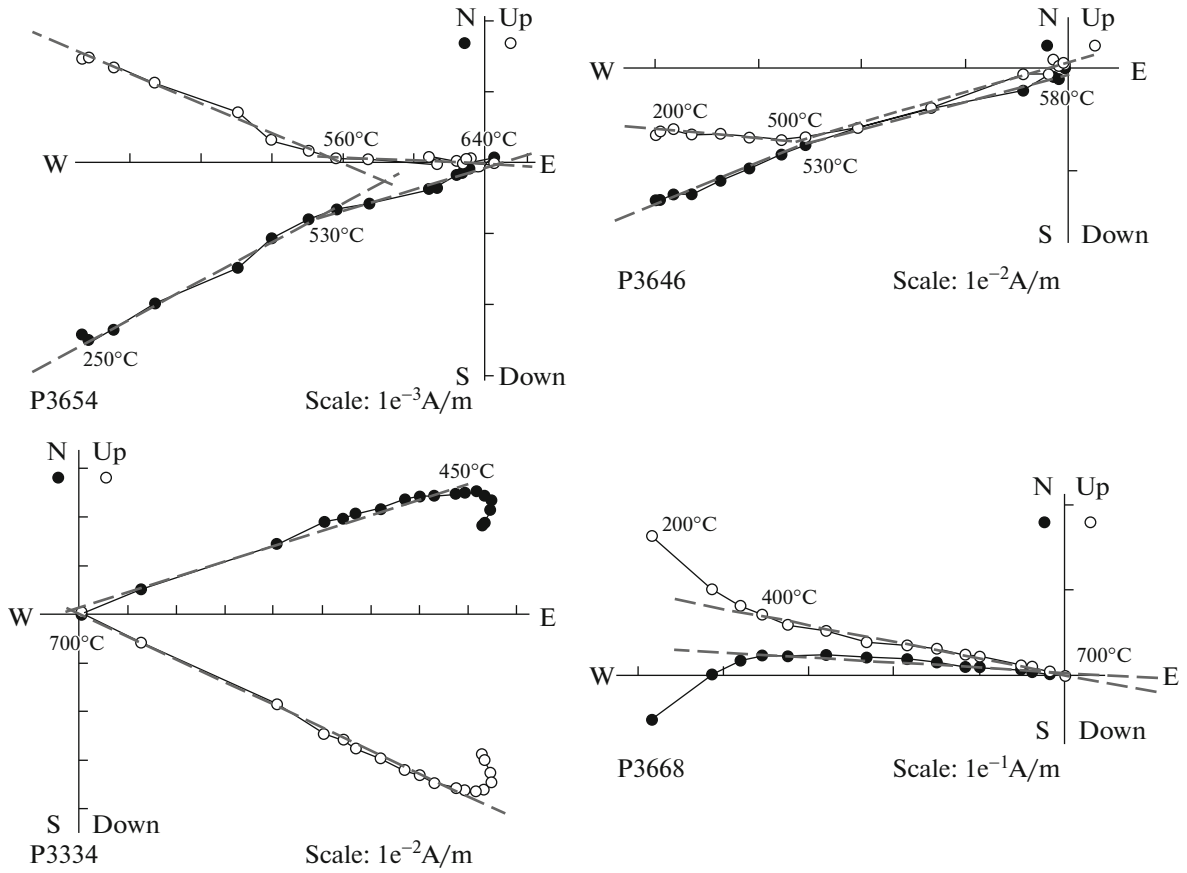


Fig. 2. Results of paleomagnetic studies. Zijderveld diagrams in the ancient coordinate system: P3654, P3646—Devonian, P3334, P3668—Lower Carboniferous. Magnetization vectors are shown by dashed lines. Open symbols are the projection of the magnetization vector J_n on the vertical plane; solid symbols are those on the horizontal plane.

magnetization was not preserved. Therefore, to determine the mutual disposition of the continent and the island arc, the data on the position of the pole of the Baltica paleocontinent from [8] were used and recalculated in the paleomagnetic directions for the continental margin (Kaga site).

The results of paleomagnetic studies (Table 1, Fig. 3) allow us to draw the following conclusions:

(1) The passive margin of the continent of Laurussia and the adjacent Magnitogorsk island arc occupied the near-equatorial position, which is confirmed by paleoclimatological studies (the occurrence of barrier reefs, etc.). The position of the continental margin and the island arc was close to sublatitudinal.

(2) The latitudinal position of the Magnitogorsk island arc in the Lower and Middle Devonian differed from the position of the continent by about 1°, and even if we take into account rather large measurement errors in the determination of the latitude, the distance between the continental margin and the island arc probably exceeded 200 km, but it was unlikely to be more than 1000 km (most likely, less).

However, it is necessary to take into account the possibility that the Kaga and Sibai points could be at different longitudes, and in order to get the current position, the collision between these tectonic structures had to be oblique at a higher subduction rate than that upon direct collision. The time of subduction of the island arc from its occurrence to the collision was approximately 30 million years. The subduction rate was likely ~3 cm/year (2.7–2.8 cm/year, according to the K.S. Ivanov's calculations [3]).

Table 1. The data obtained for high-temperature magnetization components of Devonian and Carboniferous rocks of the Southern Urals

Area	N	D	I	α_{95}	φ
Devonian, 390 Ma					
Kaga		278.3	2.4	11.0	1.2 ± 11.0
Sibai	27	253.5	0.3	6.5	0.2 ± 5.7
Lower Carboniferous, 340 Ma					
Kaga		260.1	−8.5	8.2	4.3 ± 8.2
village Ural	21	257.6	−10.9	9.0	5.5 ± 8.1

N, the number of sites; D, I (in degrees), declination and inclination in the ancient system of coordinates; α_{95}, radius of confidence circle (in degrees); φ, paleolatitude and the error of its measurement. The data are given for midpoints in the Devonian and Lower Carboniferous sections.

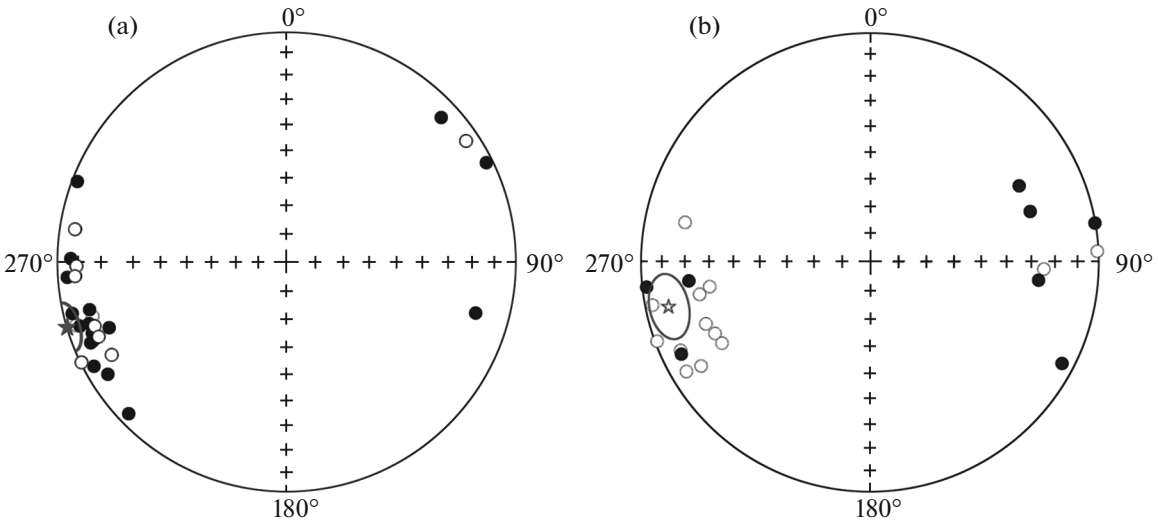


Fig. 3. Directions of high-temperature magnetization components in the Devonian (a) and Carboniferous (b) sections in the stratigraphic coordinate system. Solid symbols are projections of the magnetization vector on the lower hemisphere; open symbols are those on the upper hemisphere. A star shows general mean direction and its confidence circle.

(3) The magnetic declinations measured at the Kaga and Sibai sites differed by 25° in the Devonian; in the Lower Carboniferous, the difference was reduced to 2.5° (with minor errors).

(4) For all the differences, the data obtained for the Devonian sections by comparing our data with those from [8] are sufficiently close to the results given in [1].

(5) The declinations, inclinations, and paleolatitudes obtained for the Carboniferous sections along the profile across the Magnitogorsk Zone indicate a significant convergence of these data and their almost complete identity with those obtained for the continental margin (Kaga site).

(6) The results obtained show that the Magnitogorsk island arc in the Devonian was at some distance from the passive continental margin. The latter, during the approach, rotated counterclockwise relative to the arc by ~25°. The great similarity of inclinations and paleolatitudes of the continental margin and island arc sites indicates that, as a result of subduction and collision, the island arc and the continental passive margin collided, and thus, the passive margin was built up in the Carboniferous by accretion of the island arc. The oceanic "gap" between the continent and the island arc was transformed into the Main Ural Fault (Fig. 1).

Thus, the paleomagnetic data obtained and modern ideas about the evolution of the Uralian Orogen, in particular, the collision process between the Magnitogorsk island arc and the passive margin of the Baltica Paleocontinent (Laurussia) do not contradict each other.

ACKNOWLEDGMENTS

We lament the untimely passing of M.L. Bazhenov, who had a tremendous influence on the development of paleomagnetic studies at the Institute of Geology, Ufa Scientific Center, Russian Academy of Sciences, and, in particular, made a significant contribution to these results.

This work was supported by the Russian Foundation for Basic Research (project no. 15–05–03036) and in part by State Task no. 0252-2014-0002.

REFERENCES

1. V. S. Burtman, G. Z. Gurarii, A. V. Dvorova, et al., Geotectonics **34** (5), 397–406 (2000).
2. N. F. Danukalov, L. S. Kondruchina, and A. P. Chernikov, *Paleomagnetism of the Paleozoic of the Southern and Middle Urals* (Bashkir Branch USSR Acad. Sci., Ufa, 1983) [in Russian].
3. K. S. Ivanov, Dokl. Earth Sci. **377** (2), 164–167 (2001).
4. V. A. Maslov and O. V. Artyushkova, *Stratigraphy and Correlation of Devonian Deposits in Magnitogorsk Mega-Zone of the Southern Urals* (DizainPoligrafServis, Ufa, 2010) [in Russian].
5. G. A. Petrov and I. A. Svyazhina, Litosfera, No. 4, 23–39 (2006).
6. V. N. Puchkov, *Geology of the Urals and Cis-Urals (Actual Problems of Stratigraphy, Tectonics, Geodynamics and Metallogeny)* (DizainPoligrafServis, Ufa, 2010) [in Russian].
7. V. Bachtadse, J. Tait, H. Soffel, et al., in *Proc. Europrobe Workshop on Uralides* (Moscow, 1998), p. 23.
8. T. H. Torsvik, R. Van der Voo, U. Preeden, et al., Earth Sci. Rev. **114**, 325–368 (2012).

Translated by D. Voroshchuk